

UI/UX Design Specification: Multimodal Accessible Textbook Assistant

1. System-Wide Feature: Built-In Screen Reader Narrator

To ensure true independence for visually impaired users, the application features a proprietary, built-in screen reader. This ensures the app remains fully accessible regardless of the device's native OS accessibility settings.

1. **Continuous Text-to-Speech Conversion:** Every textual element, button, and menu item on the screen is automatically read aloud when the user's finger rests on or swipes over it.
2. **Gesture-Based Navigation:** * **Single Tap/Touch & Hold:** Reads the element under the finger.

2.1. **Double Tap:** Activates the currently highlighted element.

2.2. **Two-Finger Swipe:** Scrolls through lists or paragraphs, triggering the narrator to read the next block of text continuously.

3. **Audio Cues & Earcons:** Distinct sounds (beeps, clicks, and soft chimes) are used to indicate page loads, successful image captures, and processing states, so the user knows exactly what the application is doing without looking at the screen.

2. Introducing MARK: The Interactive Q&A Bot

MARK (Multimodal Accessible Reading Knowledge-bot) serves as the persona for the interactive question-and-answering module. Rather than a sterile search bar, MARK acts as a personalized, conversational AI tutor.

- **Voice-Activated Trigger:** Users can summon MARK at any time by tapping a persistent floating button or by using a wake word (e.g., "Hey MARK").
- **Context-Aware Memory:** MARK remembers the specific textbook page, diagram, and the user's selected age/difficulty level to provide tailored, conversational answers.

3. Updated Screen Designs

Screen 1: Splash & Profile Selection

- **Narrator Action:** Upon launch, the built-in narrator immediately announces: "Welcome to your Textbook Assistant. Swipe right to explore profiles, or double-tap anywhere to log in as a new user."
- **UI Components:**
 - **Large User Profile Cards:** High-contrast cards that read the profile name aloud upon touch.
 - **Login/Register:** Standard, large-touch-target authentication buttons.

Screen 2: Home Dashboard

- **Narrator Action:** "Home screen. The main button to scan a page is in the center of your screen."
- **UI Components:**
 - **Primary Action (Center):** A massive button labeled "**Scan or Upload Page**".
 - **Floating Action Button (Bottom Right):** A microphone icon representing **MARK**. Tapping it prompts MARK to say, "Hi, what do you want to learn today? You can say 'Scan a page' or ask me a general question."

Screen 3: Camera / Capture Interface

- **Narrator Action:** "Camera open. Move your phone slightly higher... Hold still... Page captured."
- **UI Components:**
 - **Full-Screen Viewfinder.**
 - **Bottom-Third Capture Button:** Covers the entire bottom area to ensure an easy tap.
 - **Haptic Feedback:** The phone vibrates once when the edges of the textbook page are detected, and twice when the capture is successful.

Screen 4: Configuration & Processing

- **Narrator Action:** Reads the selected age level as the user drags a slider (e.g., "Child mode selected. Processing page... Please wait.").
- **UI Components:**
 - **Age/Difficulty Selector:** Large, tactile-feeling buttons for different reading levels.
 - **Processing Graphic:** An animated, high-contrast loading ring.

Screen 5: The Learning Space (Playback & Interaction)

- **Narrator Action:** Begins reading the generated explanation using the neural TTS module.
- **UI Components:**

- **Media Controls:** Massive Play/Pause, Rewind, and Forward buttons. The built-in screen reader calls out these buttons if the user touches them, but does not interrupt the educational audio unless tapped twice.
- **Transcript Window:** Scrolling text where words highlight as they are spoken.
- **Meet MARK (FAB):** A prominent, glowing button labeled "Ask MARK."

Screen 6: Chatting with MARK (Interactive Q&A Mode)

Triggered from the Learning Space, allowing the user to interrogate the material.

1. **Narrator/MARK Voice:** "I paused the reading. What is your question?"
2. **UI Components:**

2.1. MARK Avatar/Visualizer: A friendly, high-contrast audio visualizer in the center of the screen that pulses when MARK is listening or speaking.

2.2. Hold-to-Talk Mechanism: The user holds the screen to speak their question (e.g., "MARK, what does the graph show?").

2.3. Chat History: A visual chat-bubble layout showing the user's question and MARK's generated response , which the built-in screen reader reads aloud automatically.

Screen 7: Feedback & Wrap-up

1. **Narrator Action:** "Explanation finished. Tap left if it was too hard, middle if it was good, or right if it was too easy."
2. **UI Components:**

2.1. Feedback Emojis: Three large, distinct buttons. This helps the adaptive module adjust future explanations.